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Wrist-mounted monitoring system

Abstract

A monitoring system includes a strap having a free end and a base end, a base attached to the base end of the strap, and at least one contact pad attached to the base. An electrical conductor extends along the strap between the free end and the base end. An electronic module is releasably attached to the base and includes a processor, a power supply, a transmitter, and a locking pin. A locking mechanism is configured to retain the electronic module on the base and includes a biasing member mounted on the base, a locking pawl operable between a locking position and a release position. The free end of the strap is releasably insertable between the module and the base. A key is insertable into the base and configured to release the locking pawl from the locking pin.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims the benefit of U.S. Provisional Patent application Ser. No. 63/235,249, filed on Aug. 20, 2021, which is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The invention relates to electronic personal monitoring systems that are worn by a user.

Description of the Related Art

(2) Occasionally, when a person is under arrest that person is released to “house arrest” to allow the person to live outside of the prison, but with restrictions regarding their movement ability. The person is typically required to wear some type of monitoring system that monitors the location of the person and informs authorities if the person leaves their home or other designated location. The monitoring system is typically large and can stigmatize the person when others see the monitoring system and realize that the person wearing the monitoring system is under house arrest.

(3) Also, wearers sometimes try to “escape” the monitoring by removing the monitoring system in such a way as to not alert authorities that the monitoring system has been removed.

(4) It would be beneficial to provide a monitoring system that does not look like a known monitoring system and also is difficult, if not impossible, to remove without alerting authorities that the monitoring system has been removed.

SUMMARY OF THE INVENTION

(5) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

(6) In one embodiment, the present invention is a monitoring system comprising a strap having a free end and a base end, a base attached to the base end of the strap, and at least one contact pad attached to the base. An electrical conductor extends along the strap between the free end and the base end. The electrical conductor is electrically connected to the at least one contact. An electronic module is releasably attached to the base and includes a processor, a power supply electrically connected to the electrical conductor at the base end of the strap, a transmitter, and a locking pin. A locking mechanism is configured to retain the electronic module on the base. The locking mechanism comprises a biasing member mounted on the base and a locking pawl operable between a locking position and a release position. The biasing member biases the locking pawl toward the locking position and engage the locking pin when the electronic module is attached to the base. The free end of the strap is releasably insertable between the module and the base.

(7) The present invention also provides a key that is insertable into the base and configured to release the locking pawl from the locking pin.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated herein and constitute part of this specification, illustrate the presently preferred embodiments of the invention, and, together with the general description given above and the detailed description given below, serve to explain the features of the invention. In the drawings:

(2) FIG. 1 is perspective view of a wrist-mounted monitoring system according to an exemplary embodiment of the present invention;

(3) FIG. 1A is a perspective view of the system of FIG. 1 mounted on a wrist;

(4) FIG. 1B is a schematic drawing of exemplary electronics used with the system of FIG. 1;

(5) FIG. 2 is an exploded perspective view of the system of FIG. 1;

(6) FIG. 3 is an enlarged perspective view of a base end of the system of FIG. 1;

(7) FIG. 4 is an enlarged perspective view of the base end of the system of FIG. 1, showing an electrical conductor in the strap;

(8) FIG. 5 is a top plan view of the base end of the system of FIG. 1, showing a locking mechanism engaged with a locking pin;

(9) FIG. 6 is a top plan view of the base end of the system of FIG. 5, showing a key inserted into the locking mechanism and the locking mechanism disengaged from the locking pin;
(10) FIG. 7 is a bottom perspective view of a module used with the system of FIG. 1; and
(11) FIG. 8 is a perspective view of the base end with module and the key inserted into the base.

DETAILED DESCRIPTION

(12) In the drawings, like numerals indicate like elements throughout. Certain terminology is used herein for convenience only and is not to be taken as a limitation on the present invention. The terminology includes the words specifically mentioned, derivatives thereof and words of similar import. The embodiments illustrated below are not intended to be exhaustive or to limit the invention to the precise form disclosed. These embodiments are chosen and described to best explain the principle of the invention and its application and practical use and to enable others skilled in the art to best utilize the invention.

(13) Reference herein to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment can be included in at least one embodiment of the invention. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment, nor are separate or alternative embodiments necessarily mutually exclusive of other embodiments. The same applies to the term “implementation.”

(14) As used in this application, the word “exemplary” is used herein to mean serving as an example, instance, or illustration. Any aspect or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or designs. Rather, use of the word exemplary is intended to present concepts in a concrete fashion.

(15) The word “about” is used herein to include a value of ± 10 percent of the numerical value modified by the word “about” and the word “generally” is used herein to mean “without regard to particulars or exceptions.”

(16) Additionally, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or”. That is, unless specified otherwise, or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

(17) Unless explicitly stated otherwise, each numerical value and range should be interpreted as being approximate as if the word “about” or “approximately” preceded the value of the value or range.

(18) The use of figure numbers and/or figure reference labels in the claims is intended to identify one or more possible embodiments of the claimed subject matter in order to facilitate the interpretation of the claims. Such use is not to be construed as necessarily limiting the scope of those claims to the embodiments shown in the corresponding figures.

(19) It should be understood that the steps of the exemplary methods set forth herein are not necessarily required to be performed in the order described, and the order of the steps of such methods should be understood to be merely exemplary. Likewise, additional steps may be included in such methods, and certain steps may be omitted or combined, in methods consistent with various embodiments of the present invention.

(20) Although the elements in the following method claims, if any, are recited in a particular sequence with corresponding labeling, unless the claim recitations otherwise imply a particular sequence for implementing some or all of those elements, those elements are not necessarily intended to be limited to being implemented in that particular sequence.

(21) The wrist-mounted monitoring system according to the present invention is similar to known ankle-mounted monitoring systems that are used to monitor the location and movement of the

wearer, such as those under house arrest or are otherwise subject to official monitoring, but does not include the stigma associated with wearing the ankle-mounted monitoring system.

(22) Referring to the Figures, a wrist-mounted monitoring system **100** according to an exemplary embodiment of the present invention is shown in FIG. **1** and includes a wrist strap **110** and a module **130** that houses operational electronics such as a CPU **139** (shown in FIG. **1B**) and a locking/unlocking mechanism **132** in a housing **134**. Locking/unlocking mechanism **132** requires the use of a special key **150** to release locking/unlocking mechanism **132** and allow strap **110** to be removed from the wrist **52** of a wearer **50**. FIG. **1A** shows the system attached to wrist **52** of wearer **50**.

(23) Referring to FIG. **1B**, monitoring system **100** wirelessly communicates with a central monitoring system **60** through a receiver/transmitter **62** to monitor the location of monitoring system **100**.

(24) Referring to FIGS. **1-4**, strap **110** is constructed from a generally planar elongate flexible material such as, for example, a polymer. Strap **110** includes a first edge **112**, an opposing second edge **114**, and a plurality of transversely extending generally oval through-openings **115** formed between first edge **112** and second edge **114** and evenly spaced along a length of strap **110**.

(25) A base **118** for module **130** is located at a base end **120** of strap **110** and a free end **121** of strap **110** is located distal from base end **120**. Base **118** includes a platform **119** having openings for a plurality of raised oval tabs **122** spaced to allow insertion of tabs **122** into a like plurality of through-openings **115** when strap **110** is wrapped around the wrist **52** of wearer **50** and monitoring system **100** is secured to the wrist **52**. Tabs **122** extend upwardly from base **118**, as shown in FIG. **4**. Base **118** also includes a pair of slots **123** distal from strap **110** that allow key **150** to be inserted therein to remove module **130** from base **118**. The entrance to slots **123** can be covered with removable rubber inserts **125** to prevent water an/or dirt from entering into slots **123** and possibly interfering with the operation of key **150** when key **150** is inserted into slots **123**.

(26) Strap **110** also includes a conductor **126** extending along a length of strap **110**. Conductor **126** is used to complete an electrical circuit when monitoring system **100** is wrapped around wrist **52** of wearer **50** so that, if strap **110** is cut, the electrical circuit is also cut. Conductor **126** has a base end **127** attached to base **118** and electrically connected to contact pads **131** in base **118**. Contact pads **131** will provide for an electrical connection to module **130** when module **130** is attached to base **118**.

(27) A strap retainer **129** is slidably mounted on strap **110**. Strap retainer **129** is used to retain free end **132** when strap **110** is wrapped around wrist **52**. Strap retainer **129** is slidable to accommodate different wrist thicknesses.

(28) Base **118** is a generally planar, oval shaped base constructed from a rigid material, such as polycarbonate/nylon. Base **118** supports and retains module **130**. Module **130** houses all electronics, such as CPU **139**, wireless transmitter **133** (to transmit location and other information to monitoring system **60**), and other electronic components **135**, including a battery **137**, that are used to operate monitoring system **100**, shown in FIG. **1B**. Module **130** also houses a locking mechanism **132** that releasably retains module **130** onto base **118**.

(29) Referring to FIGS. **5** and **6**, locking mechanism **132** includes a single biasing member **134** in the form of a torsion spring that is located below platform **119**. Biasing member **134** is a single piece torsion spring with a first leg **136** and a second leg **138** connected at an acute angle at an apex **140**. First leg **136** and second leg **138** are biased away from each other. Each leg **136**, **138** is in contact with a locking pawl **142**, **144**, respectively, that has a pivot end **146**, **148**, respectively, and an engaging end **162**, **164**, respectively.

(30) Engaging ends **162**, **164** are biased by biasing member **134** away from each other and into engagement with locking pins **170**, **172**, respectively, when module **130** is attached to base **118** (Locking pins **170**, **172** are shown in FIG. **7**). Each locking pin **170**, **172** has a recessed portion **174**, **176**, that receives an engaging end **162**, **164** so that module **130** is secured to base **118**, with

strap **110** secured in between module **130** and base **132** around wrist **52** so that monitoring system **100** cannot be removed from wrist **52** without using key **150**. Locking pins **170**, **172** fit into pin holes **124** in base **118** to secure locking pins **170**, **172** with base **118**.

(31) Referring to FIG. 7, a pair of pogo pins **178**, **180** extend from module **130** and are used to engage conductor **126**. When module **130** is fixed over base **118**, pogo pins **178**, **180** engage contact pads **131** in module **130** such that pogo pins **178**, **180** electrically connect conductor **126** to CPU **139**.

(32) To secure monitoring system **100** around a wrist **52**, wrist **52** is held out, palm side down. Base **118** is placed on top of wrist **52** and strap **110** is wrapped around wrist **52**. Through openings **115** in strap **110** are aligned with tabs **122** on base **118** to secure strap **110** to base **118**. Module **130** is then snapped onto base **118** such that locking pins **170**, **172** engage their respective locking pawls **142**, **144** to secure module **130** onto base **118**.

(33) Contact is made by pogo pins **178**, **180** when free end **121** of strap is placed over base **118** and module **130** is snapped onto base **118**. An electrical circuit is formed from battery **137**, to CPU **139**, to base end **127** of conductor **126**, to free end **121** of conductor **126**, to pogo pins **178**, **180**, and back to battery **137**. When the electrical circuit is complete, transmitter **133** transmits to receiver **62**, indicating that system **100** is secured to the wearer **50** and that the location of the system **100** and the wearer **50** are known. When the electrical circuit is broken, either by removing module **130** from base **118**, by cutting strap **110**, or otherwise, transmission from transmitter **133** to receiver **62** stops and an alarm can be sounded via monitoring system **60** that system **100** is no longer being worn by the wearer **50**, and that the location of wearer **50** is now in doubt.

(34) Key **150** is shown in FIGS. 6 and 8. Key **150** is used to release locking mechanism **132** so that strap **110** can be loosened from base **118** and removed from wrist **52**. Key **150** is generally U-shaped and includes a handle **152** with two parallel extending legs **154**, **156**.

(35) Each leg **154**, **156** has a tapered tip **158**, **160**, respectively, with each taper facing the opposing leg **156**, **154**. To use key **150**, legs **154**, **156** are inserted into release openings **123** and advanced into base **118** until tips **158**, **160** engage pawls **142**, **144** and bias engaging ends **162**, **164** of pawls **142**, **144** toward each other, moving pawls **142**, **144** out of engagement with locking pins **170**, **172**. When pawls **142**, **144** are disengaged from locking pins **170**, **172**, module **130** can be removed from base **118**. With module **130** removed from base **118**, free end **121** of strap **110** can be removed from base **118** and strap **110** can be removed from wrist **52**. Monitoring system **100** is configured such that free end **121** of strap **110** is configured to be removed from base **118** only when electronic module **130** is removed from base **118**.

(36) It will be further understood that various changes in the details, materials, and arrangements of the parts which have been described and illustrated in order to explain the nature of this invention may be made by those skilled in the art without departing from the scope of the invention as expressed in the following claims.

Claims

1. A monitoring system comprising: a strap having a free end and a base end; a base attached to the base end of the strap; at least one contact pad attached to the base; an electrical conductor extending along the strap between the free end and the base end, wherein the electrical conductor is electrically connected to the at least one contact; an electronic module releasably attached to the base, the electronic module comprising: a processor; a power supply electrically connected to the electrical conductor at the base end of the strap; a transmitter; and a locking pin; and a locking mechanism configured to retain the electronic module on the base, the locking mechanism comprising: a biasing member mounted on the base; a locking pawl operable between a locking position and a release position, the biasing member biasing the locking pawl toward the locking position and engaging the locking pin when the electronic module is attached to the base; wherein

the free end of the strap is releasably insertable between the module and the base.

2. The monitoring system according to claim 1, wherein the electronic module further comprises a pogo pin extending therefrom, the pogo pin configured to engage the electrical conductor.

3. The monitoring system according to claim 2, wherein, when the free end of the strap is inserted into the base, an electrical circuit is formed from the power supply to the processor to the electrical conductor at the base end of the strap, through the electrical conductor to the free end of the strap, to the pogo pin, and back to the power supply.

4. The monitoring system according to claim 1, further comprising a key insertable into the base and configured to release the locking pawl from the locking pin.

5. The monitoring system according to claim 4, wherein the key comprises a leg configured to engage the locking pawl and bias the locking pawl away from the locking pin.

6. The monitoring system according to claim 4, wherein the locking pawl is mounted on the base.

7. The monitoring system according to claim 1, wherein the biasing member comprises a torsion spring.

8. The monitoring system according to claim 7, wherein the torsion spring comprises a first leg and a second leg connected to each other at an acute angle at an apex.

9. The monitoring system according to claim 1, wherein the free end of the strap is configured to be removed from the base only when the electronic module is removed from the base.

10. The monitoring system according to claim 1, wherein the strap includes at least one through opening at the free end and wherein the base comprises a tab configured to be inserted into the at least one through opening such that, when the electronic module is attached to the base, the free end of the strap is secured between the electronic module and the base.

11. A monitoring system comprising: a strap having a free end and a base end; a base attached to the base end of the strap; at least one contact pad attached to the base; an electrical conductor extending along the strap between the free end and the base end, wherein the electrical conductor is electrically connected to the at least one contact; an electronic module releasably attached to the base, the electronic module comprising: a processor; a power supply electrically connected to the electrical conductor at the base end of the strap; a transmitter; a pogo pin extending therefrom, the pogo pin configured to engage the electrical conductor; and a locking pin; and a locking mechanism configured to retain the electronic module on the base, the locking mechanism comprising: a biasing member mounted on the base; a locking pawl mounted on the base and operable between a locking position and a release position, the biasing member biasing the locking pawl toward the locking position and engaging the locking pin when the electronic module is attached to the base; wherein the free end of the strap is releasably insertable between the module and the base; and a key insertable into the base and configured to release the locking pawl from the locking pin, the key comprising: a leg configured to engage the locking pawl and bias the locking pawl away from the locking pin.

12. The monitoring system according to claim 11, wherein the system is configured to be worn on a wrist.

13. The monitoring system according to claim 11, further comprising a receiver configured to receive electronic signals from the transmitter.

14. The monitoring system according to claim 13, wherein, when the electronic module is attached to the base, an electrical circuit is formed along the strap and the electronic signals are transmitted from the transmitter to the receiver.

15. The monitoring system according to claim 14, wherein, when the electronic module is not attached to the base, the electrical circuit is broken and the electronic signals are not transmitted from the transmitter to the receiver.

16. The monitoring system according to claim 14, wherein, when the strap is cut, the electrical circuit is broken and electronic signals are not transmitted from the transmitter to the receiver.

17. The monitoring system according to claim 11, wherein the locking pin has a recessed portion

configured to receive an engaging end of the locking pawl.

18. The monitoring system according to claim 11, wherein the free end of the strap is configured to be removed from the base only when the electronic module is removed from the base.

19. The monitoring system according to claim 11, wherein the key has a first leg and a second leg configured to be inserted into the base.

20. The monitoring system according to claim 11, wherein, when the free end of the strap is inserted into the base, an electrical circuit is formed from the power supply to the processor to the electrical conductor at the base end of the strap, through the electrical conductor to the free end of the strap, to the pogo pin, and back to the power supply.
